Module Code: CO832

Examination number: 199662

University of Kent

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Examination Answer Booklet

Before starting this examination, you **MUST enter your Examination Number** above **(double click)**; failure to provide this means your mark cannot be recorded and **you may be awarded a 0 (zero)** for this examination.

Re-save the answer booklet to include your exam number in the file name, using the following format: [**your exam number]** [**module code] AnswerBooklet** (e.g. *123456 XX123 AnswerBooklet*).

Answer one question per page. You may add additional pages to this document if necessary.

Question Number	1a

Add your answer below:

ID	Age	Salary	Tax Paid	Tax Debt
001	18	£25000	Yes	£0
002	25	£35000	No	£3000
003	42	£30000	No	£1500
004	23	£20000	Yes	£0
005	36	£40000	Yes	£0
006	56	£22000	No	£600
007	43	£29000	No	£2500
008	19	£19000	Yes	£0
009	23	£29000	Yes	£0
010	24	£25000	Yes	£0
011	43	£60000	No	£
012	37	£75000	Yes	£0
013	27	£23000	Yes	£0
014	25	£20000	No	£1000
015	28	£30000	Yes	£0
016	25	£28000	No	£250
017	23	£24000	Yes	£0
018	35	£100000	Yes	£0
019	40	£150000	No	£50000
020	20	£30000	Yes	£0

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Question Number	1b

Add your answer below:

P (< Fever = “High”, New Cough = “No” >) = 4/14

No is the class attribute.

P (< Fever = “High”, New Cough = “Yes” >) = 1/14

Yes is the class attribute.

P (< Fever = “No”, New Cough = “No” >) = 3/14

No is the class attribute.

Question Number	1c

Add your answer below:

i)

Gender x Buy

	yes	no
M	2	3
F	2	3
Other	2	3

Salary x Buy

	yes	no
High	4	1
Medium	3	2
Low	0	5

ii)

Gender will have high conditional entropy as it is less predictable due to equal distribution throughout. Salary will have low conditional entropy as the distribution varies meaning more chance of predictability.

Question Number	3a

Add your answer below:

PCA reduces redundant features from this data set, which would make the decision tree algorithm learn at a much more efficient speed. PC 1,2,3 and 4 could be used to

learn, as it covers 90% of the variance within the dataset. The random forest algorithm is also another algorithm that has the same benefits from using PCA sets.

Question Number	3b

Add your answer below:

The error with the algorithm, is that the researcher has picked the highest value from running the algorithm 10 times, without checking for cross-validation. This is an issue because it is inaccurate, especially with using the same testing sets.

Question Number	3c

Add your answer below:

i) Classification
ii) Clustering
iii) Classification
iv) Discovery of Association Rules

Question Number	4a

Add your answer below:

Since there are 3 correlated labels and 1 uncorrelated label, Both methods should be used for this.

Patient Id	Gender	Age	Dementia?

Patient Id	Gender	Age	Diabetes?

Patient Id	Gender	Age	Predictive Diabetes?	Heart Disease?

Patient Id	Gender	Age	Predictive Diabetes?	Predictive Heart Disease?	Lung Cancer?

The first method, binary split, was done because you split them individually. The second method, chain link, was used because it will take the 3 correlated labels to make the other 3 tables. They are in order of easiest predictability.

Question Number	4b

Add your answer below:

i) Powerset method, because you can create compound labels which can combine labels in the original data set. This is to help with labelling similar values together.

ii)

Patient id	Gender	Age	Class
1	M	18	Dementia and Heart Disease
2	M	25	Lung Cancer
3	F	20	Dementia and Heart Disease
4	F	22	Lung Cancer
5	M	30	Dementia and Heart Disease
6	M	28	Lung Cancer
7	M	27	Dementia and Heart Disease
8	F	24	Lung Cancer
9	F	19	Dementia and Heart Disease
10	F	21	Lung Cancer

Question Number	4c

Add your answer below:

i) I think rule r2 is not more useful than rule r1 because I do not think the class attribute of if statement in rule r2 makes it a greater probability of the instance satisfying the antecedent of rule r1. The reason for this is that rule r1 is more inclusive than r2, increasing the likelihood of finding more customers in r1 than r2 (since r2 is excluding more customers).

ii) I think rule r3 is most likely to be more interesting. The reason for this is that the extra predictor attribute and class attribute will increase the chances of representing new patterns. As it contains more information.